

The FOAF profile about a person can be created using the URL <http://www.ldodds.com/foaf/foaf-a-matic.html>.

The FOAF file will be created with RDF as follows:

```
<rdf:RDF
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
  xmlns:foaf="http://xmlns.com/foaf/0.1/"
  xmlns:admin="http://webns.net/mvcb/">
<foaf:PersonalProfileDocument
  rdf:about="">
  <foaf:makes rdf:resource="#me"/>
  <foaf:primaryTopic
  rdf:resource="#me"/>
  <admin:generatorAgent
  rdf:resource="http://www.ldodds.com/foaf/foaf-a-matic"/>
  <admin:errorReportsTo
  rdf:resource="mailto:leigh@ldodds.com"/>
</foaf:PersonalProfileDocument>
<foaf:Person rdf:ID="me">
  <foaf:name>Gaurav Kumar</foaf:name>
  <foaf:title>Mr</foaf:title>
  <foaf:givenname>Gaurav</foaf:givenname>
  <foaf:family_name>Kumar</foaf:family_name>
  <foaf:nick>GK</foaf:nick>
  <foaf:mbox_sha1sum>eccb3d9dd0c3a386706462b4288ac0ff2678ab4</foaf:mbox_sha1sum>
  <foaf:homepage
  rdf:resource="gauravkumarindia.com"/>
  <foaf:knows>
  <foaf:Person>
  <foaf:name>friend1</foaf:name>
  <foaf:mbox_sha1sum>4d2e2bea6b04249b226fe9b1761b32554f19e55d</foaf:mbox_sha1sum>
  </foaf:Person></foaf:knows>
  <foaf:knows>
```

```
<foaf:Person>
<foaf:name>friend2</foaf:name>
<foaf:mbox_sha1sum>c666c29d65fcd3461f8f81d3bc76e25802893e9f</foaf:mbox_sha1sum>
</foaf:Person></foaf:knows></foaf:Person>
</rdf:RDF>
```

The FOAF file can be used as the database and matched with the other profiles using semantic Web protocols and algorithms.

The FOAF search engine (<https://www.foaf-search.net/>) is available and can be used to extract information about any person (Figure 1).

Open source tools and libraries for the semantic Web include:

- Apache TinkerPop: <https://tinkerpop.apache.org>
- RDFLib: <https://github.com/RDFLib/rdfli>
- Apache Jena: <http://jena.apache.org/>
- Protégé: <https://protege.stanford.edu>
- Sesame: <http://www.openrdf.org/>
- Linked Media Framework: <https://code.google.com/p/lmf/>
- Open Semantic Framework: <http://opensemanticframework.org/>
- D2R: <http://d2rq.org/d2r-server>
- Paget: <http://code.google.com/p/paget/>
- Semantic Media: http://semantic-mediawiki.org/wiki/Semantic_MediaWiki

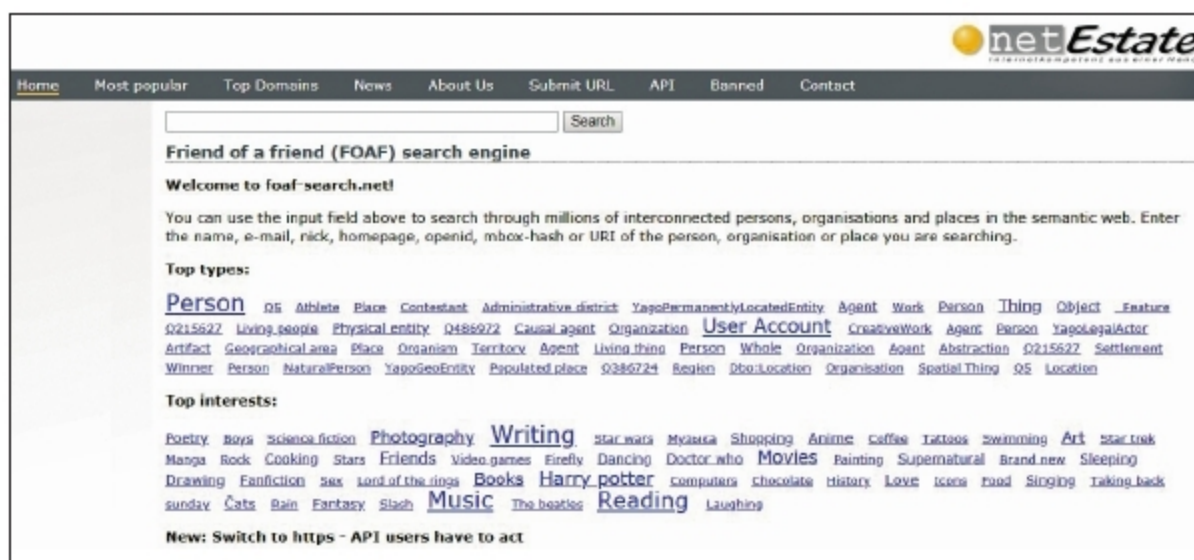


Figure 1: FOAF search engine

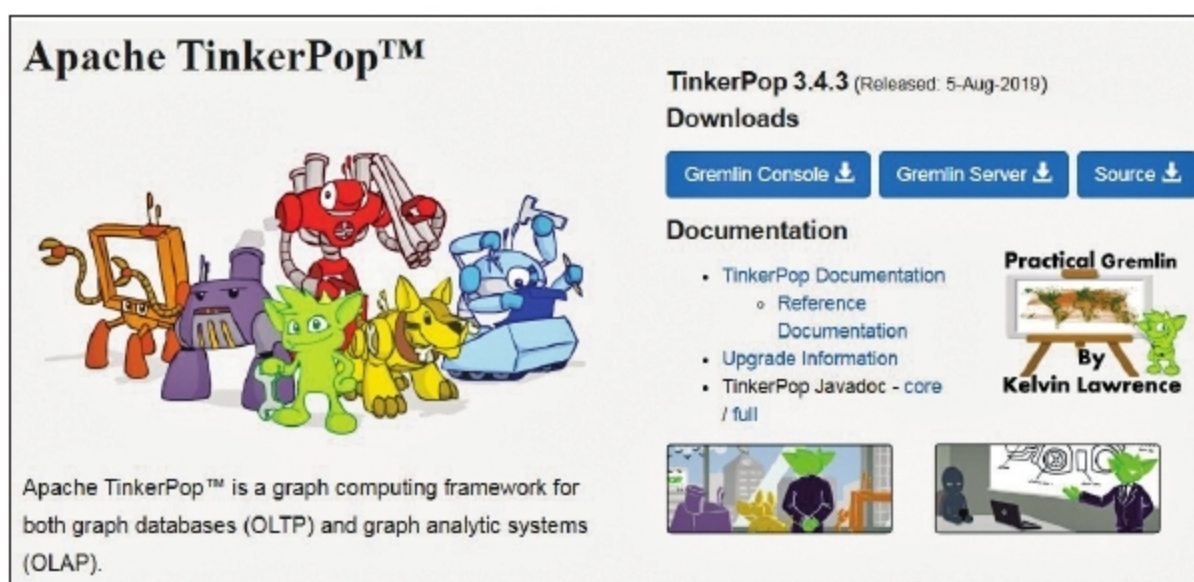


Figure 2: Official portal of Apache TinkerPop

Installing Apache TinkerPop on Windows

Apache TinkerPop is one of the multi-featured and powerful frameworks for the development and programming of semantic Web applications. Available as a free and open source distribution, it is primarily used for graph computing and ontology based programming, which is quite useful in online transaction processing and graph analysis in social media based applications.

The dynamic graph of people, and their relationships/relevance to a particular topic, can be created. These graphs are used for the predictive mining of relationships in social media, as shown in Figures 3 and 4.